

Appendix T

Detailed Thematic Analysis of the Perceived Value of LitTraceAI in Research Practice

T.0.1 AI as a Drafting Tool

This theme captures how experts perceived LitTraceAI as a supportive resource for producing early drafts of the literature review section and accelerating its related activities. Experts generally viewed the tool as useful for initiating the writing process while still requiring human refinement and critical review.

Regarding the generation of an initial draft, 11 out of 15 experts highlighted the value of having a starting structure for developing the final text. Expert 16 stated that LitTraceAI “*provides a solid starting point to build from,*” while Expert 15 described LitTraceAI as “*a good start to have as a template for the paper.*” This perception was also reflected by Expert 6:

It [LitTraceAI] gives me a first glimpse of the literature review and from there I can build a final version. Also, sometimes we researchers waste too much time

finding discourse markers, and with this AI we avoid losing time on matters of form. It is a very powerful tool for taking its content as a starting point to build the final version.

Together, these responses indicate that experts perceived the generated content as a scaffold that could support the early stages of writing by providing structure, organization, and a preliminary synthesis of the literature.

At the same time, experts consistently emphasised the importance of researcher oversight and validation. Expert 9 stated:

In the end, as a researcher, it is up to you to curate what the tool [LitTraceAI] can give you, since you will take whatever is most appropriate for the paper you will write. In other words, it would be a good draft that requires the researcher’s judgment and expertise for its improvement.

Likewise, Expert 12 explained that LitTraceAI could be used “*as a first draft,*” but that the researcher would “*have to improve it later and confirm that the information is correct.*” These accounts suggest that experts did not perceive the AI-generated output as ready for direct adoption, but rather as material requiring verification, adaptation, and researcher judgment before being incorporated into academic writing.

This finding aligns with Natarajan et al. (2025) conception of **AI-in-the-Loop (AI2L)** systems. Natarajan et al. (2025) argue that the key difference between Human-in-the-Loop (HIL) and AI-in-the-Loop (AI2L) systems lies in control and accountability. Whereas HIL systems remain largely AI-centred, with the model driving the workflow and humans providing corrective input, AI2L systems are expert-centred. In AI2L systems, human experts retain responsibility for decision-making from start to finish, while AI serves a supportive role by generating information, suggestions, or analyses that inform, but do not replace, expert judgement.

Experts consistently described LitTraceAI as a tool for generating initial drafts rather than producing outputs suitable for direct adoption. The need for verification, adaptation, and critical evaluation indicates that experts did not regard the system as an autonomous producer of academic content. Instead, they viewed its outputs as provisional resources requiring expert judgement before incorporation into scholarly writing.

This interpretation is consistent with Natarajan et al. (2025) argument that **AI2L** systems are particularly appropriate for complex, contextual, and high-stakes tasks, where expert knowledge and situational judgement remain essential. Whereas well-defined and narrowly bounded tasks may be suitable for greater automation, academic writing and literature synthesis involve interpretation, evaluation of evidence, contextual understanding, and responsibility for the accuracy of claims. As such, the role of the expert cannot be reduced to merely validating AI-generated outputs. Rather, LitTraceAI appears to function as a collaborative decision-support tool that augments scholarly expertise while leaving control, accountability, and final judgement firmly in the hands of the researcher.

In addition to supporting drafting activities, 4 out of 15 experts associated LitTraceAI with increased efficiency in writing. Expert 4 explained that LitTraceAI would help “*compile state of the art sections more quickly,*” although the generated content would still function as “*an initial draft.*” Similarly, Expert 9 highlighted the potential reduction in workload:

I would use it because it helps you gain speed. What used to take a month, such as producing a state of the art, can perhaps now be done in a week, thanks to this tool [LitTraceAI]. On top of that, the citations are reliable and it does not hallucinate.

These responses indicate that experts perceived LitTraceAI as assisting the writing workflow by accelerating preliminary synthesis and drafting activities. This finding aligns with the concept of **efficiency gains**. As described through Expert 3’s observation in Section 7.1.2, LitTraceAI’s advantage lies in its ability to process and synthesise large volumes of information more efficiently than manual analysis. Accordingly, the reported reductions in workload

and faster production of state-of-the-art sections suggest that the perceived benefits arise from efficiency gains attributable to the technology itself.

Although experts generally emphasised the usefulness of LitTraceAI, some concerns regarding responsible use also emerged. Expert 17 expressed uncertainty about whether partial use of AI-generated text *“would be considered plagiarism.”* This concern may indicate that, despite recognising the drafting benefits of LitTraceAI, certain experts remained cautious about the ethical implications associated with integrating AI-generated content into academic writing. As discussed in Section 4.1.2, the tool is intended to support researchers by providing an initial draft that serves as a stimulus for reflection and critical evaluation, rather than finalised scholarly text. Researchers remain responsible for questioning, revising, and improving the generated content.

T.0.2 AI as a Cognitive Support for Literature Synthesis

Experts perceived LitTraceAI as a supportive resource for synthesising and making sense of the literature. Across responses, LitTraceAI was associated with helping researchers identify gaps, compare studies, organise information, recognise themes, and surface relevant papers. At the same time, experts consistently emphasised that the generated outputs still required researcher judgment, verification, and refinement.

The identification of research gaps (7/18) was one of the most-frequently discussed capabilities. Experts described LitTraceAI as useful for highlighting limitations and underdeveloped areas in the literature that could support exploratory thinking and the early definition of research directions. Expert 18 explained that the tool could be useful *“to identify research opportunities for a construct that is still under development”*. Similarly, Expert 10 stated that *“it helps you see the limitations and the gaps that exist on the topics”*.

This perception was reinforced by Expert 22, who noted that LitTraceAI would be valuable *“to give me ideas ... about possible gaps. In that sense it is better than ChatGPT. However, I would have to go further to verify whether it is really a gap or not.”* Likewise,

Expert 8 explained that LitTraceAI could help identify “*unsupported limitations in the analysis*,” but that after identifying “*the gaps and limitations*,” the expert would still need “*to search for support for these limitations*”. Together, these responses suggest that experts valued LitTraceAI as a means of generating and exploring potential research directions through the identification of gaps and limitations in the literature.

Comparative support (5/18) also emerged as an important perceived value of LitTraceAI, particularly for discussion and synthesis activities. Expert 13 explained:

Yes, for the discussion. The material or information provided should help you not only compare similarities, but also discrepancies. In any case, it [LitTraceAI] should help you contrast the similarities and differences between your results and published findings.

Similarly, Expert 19 stated that LitTraceAI could support findings “*either in favour or against*” his own analysis because it “*provides both types of information*.” These responses indicate that experts perceived LitTraceAI as supporting analytical comparison across studies rather than merely summarising content.

A more detailed example was provided by Expert 1, who described using LitTraceAI during the initial stages of the literature review process:

Yes, for example, I would use it in the initial stages. At the beginning, when I search for my papers, I select them, upload them to the platform, and ask it to help me organise comparative structures, contrasts, and limitations. Obviously, before that, I have already pre-selected and read the papers, because that is the process we normally follow. These are not unknown papers to me; I have already reviewed them beforehand. Then I upload them to the platform, review the comparisons, limitations, and contrasts, and evaluate whether I agree or disagree with the authors on those points. There may be differences or similarities, but

the tool [LitTraceAI] greatly facilitates the development of this synthesis. Being able to do this seems remarkable to me.

This example illustrates how experts used LitTraceAI to facilitate the comparison, contrast, and synthesis of findings across multiple studies.

Experts also associated LitTraceAI with organising and structuring literature (3/18). Expert 22 stated that *“it would be valuable ... in helping me summarise, structure, and give me ideas about organisation.”* Similarly, Expert 27 emphasised the value of assembling and organising literature relevant information rather than fully generating the review itself:

I think this is where that middle option comes in, what I like is that it’s pulling out things like: here are sentences introducing the topic, here are sentences contrasting ideas, here are sentences about findings, right? I mean, those are things you’d normally have to do yourself, and I don’t really want the literature review generated so much as brought together.

These responses indicate that some experts perceived value in LitTraceAI as a mechanism for structuring and organising information to facilitate synthesis, rather than as a tool that autonomously writes the final literature review.

Less frequently, experts referred to the identification of major themes (2/18) and the discovery of overlooked papers (1/18). Expert 16 mentioned that *“it’s helpful that it identifies the major themes”* for inclusion in a literature review. Similarly, Expert 3 explained that LitTraceAI could help *“discover maybe papers that maybe I’ve been ignoring for some reason.”*

This perception was also reflected by Expert 27, who highlighted the potential of grouping information across multiple studies:

There’s also a use case for bringing together sentences from multiple papers. Can we read all the papers at once? Well, no. But if you shuffle in the literature review

relevant sentences and then group and cluster them, I think there's something with huge potential there.

Together, these responses highlight the potential of LitTraceAI to support thematic exploration and increase awareness of relevant literature that might otherwise be overlooked.

Across these examples, experts consistently described LitTraceAI as a cognitive support tool that assisted with identifying gaps, comparing studies, organising information, and exploring themes within the literature. Rather than replacing scholarly analysis, the tool was viewed as augmenting researchers' ability to synthesise evidence and navigate large bodies of literature. This perspective aligns with the concept of **AI-in-the-Loop**, whereby AI contributes analytical support while researchers retain responsibility for interpretation, evaluation, and decision-making.

Although experts generally perceived value in LitTraceAI as an assistive resource for literature synthesis and exploration, some also expressed clear boundaries regarding authorship and intellectual responsibility. For example, Expert 3 stated that *"I don't see myself using a full AI text because I can't claim it is my work."* This concern is consistent with the principles outlined in Section 4.1.2, which position LitTraceAI as a tool for supporting rather than replacing scholarly writing.

T.0.3 AI Supporting Research Training

This theme reflects how experts perceived LitTraceAI as a supportive resource for research training by facilitating students' engagement with literature analysis while also requiring active instructor mediation and critical oversight.

Learning support (6/9) was reflected in experts' descriptions of LitTraceAI as a mechanism that could help students develop analytical and reflective skills. Expert 24 explained that the tool could be used *"so that they [students] evaluate the truthfulness of the content and develop critical thinking and deeper reading skills."* Similarly, Expert 21 described the tool as useful for helping students *"quickly make a comparison of trends and other criteria,"*

while emphasising that the learning process would involve jointly analysing “*what is right and what is wrong in the tool’s response.*” These statements suggest that experts valued LitTraceAI not simply for generating content, but for supporting to some extent students’ active engagement with literature interpretation.

Expert 1 also described LitTraceAI as valuable for postgraduate and doctoral students because conducting a thesis requires students to engage deeply with a topic. The expert explained that “*this tool encourages critical thinking if you correctly analyse its content and evaluate what makes sense and what does not.*” This perspective further indicates that experts perceived the educational value of the tool as dependent on students examining critically the generated content rather than accepting it passively.

Expert 25 also emphasised the importance of reflective evaluation:

This tool [LitTraceAI] is useful for my doctoral students. The process would be that students would screen the papers and then upload them to this tool. Then they must explain to me what is right and what is wrong. Based on this, they should realise what is missing in their theoretical framework and what they are doing correctly.

This quote may indicate that experts viewed the generated content as a starting point for discussion and self assessment rather than as a completed text that should be accepted without revision.

Pedagogical mediation (3/9) was reflected in how experts described the importance of teaching, instructor guidance, and discussion when using LitTraceAI with students. Expert 8 explained that students who are beginning to write papers should “*analyse the content of the tool*” to determine “*what is right and what is wrong,*” adding that “*if it is taught correctly, critical thinking can be developed in students*”. This suggests that experts perceived the educational value of LitTraceAI as dependent on active teaching practices and guided reflection rather than independent use by students.

This teaching oriented perspective was also reflected by Expert 2, who stated: *“I am sure that my colleague in Argentina, who is the leader of the research group, would be interested in using this tool [LitTraceAI] with his students or postdocs.”* Although brief, this statement may indicate that some experts perceived LitTraceAI as having potential value within supervised research training environments involving both students and early career researchers.

Expert 26 expanded this teaching perspective by arguing that errors in AI-generated content could themselves become pedagogically valuable:

AI [LitTraceAI] does not have to be perfect. In fact, it is important that it makes mistakes so that the student can detect the error and correct it. Otherwise, Kahneman’s economic psychology is activated, according to which people tend to use System 1, which is fast and intuitive, when something seems easy or convenient, instead of analysing it deeply and rationally with System 2.

This perspective suggests that some experts perceived imperfection as part of the educational value that LitTraceAI should have. In this interpretation, an overly fluent or convenient AI output may encourage students to rely on System 1, accepting the generated content quickly and intuitively. By contrast, visible errors may encourage students to activate System 2 by engaging in slower, more deliberate, and critical evaluation. Thus, the findings may suggest that some experts perceived LitTraceAI’s outputs as potentially useful for fostering critical engagement with literature in research training contexts when accompanied by questioning, discussion, and instructor guidance.

Taken together, these accounts suggest that experts perceived LitTraceAI as a resource that could encourage critical engagement with literature by prompting students to evaluate, question, and discuss AI-generated outputs. Rather than functioning as an autonomous teaching system, the tool was viewed as supporting research training activities when accompanied by instructor guidance and reflective analysis. This perspective aligns with the concept of **AI-in-the-Loop**, whereby AI contributes analytical input while students and

educators retain responsibility for critical evaluation, interpretation, and knowledge construction.